BOREL CAPITAL · RESEARCH & TECHNICAL SPECIFICATION

A multi-layer alternative-data research system

First application, I1: crude inventories at Cushing before the weekly EIA report. Physical measurement as the central thesis (B), a published price effect as a separate replication (A).

Hamza Khan · version 1.2 (revision 3) · 26 October 2026 · companion to the application deck (45 pages)
Repository `Multilayer_AltData` (github.com/HamzaKhan2002) · pre-registrations v1 `66bec92f…cbbe`, v2 `acfe66ef…d84f` and v3 `d0b8d4c6…4ec9`

This document complements the deck. The deck gives the argument; this document gives a quantitative researcher what is needed to judge it: definitions, models, protocols, the exact files behind each number, and the statistical limits of what has been and can be shown. Paths are relative to the repository root. No strategy backtest, paper record or live result exists, and none is implied.

IMPLEMENTED	Code in the repository that runs and is tested; no claim about markets.
EMPIRICAL FINDING	A number computed by a script of the repository, with its output file; pre-registered unless marked exploratory.
EXPLORATORY	Computed, but a validity rule was amended after a first run, or no decision rule was fixed in advance.
PROPOSED	A protocol or design that has not been run. Never presented as a result.
ASSUMPTION	An input that is chosen, not measured: costs, latency, capacity, coverage.
ILLUSTRATIVE	Arithmetic on assumptions (PnL grids, break-even examples). Never a result.

1 Executive summary

Objective. Build, within a six-month internship, one systematic strategy whose edge comes from information the market has not yet priced, and prove it at a level that justifies capital. The system is asset-class independent: sources, measurement and regimes change per idea; point-in-time discipline, validation and risk do not. Six ideas share it; I1 is prototyped first (a free version is built and tested) because its data can be tested now on free sources.

> **The decision this document supports.** Fund a bounded experiment on professional Cushing data. Three decisions need three levels of evidence: **fund an experiment** (a documented mechanism, an uncertainty the experiment can resolve, a bounded cost and scope: this request); **subscribe for a strategy** (incremental value measured on that source's own data); **commit market risk** (independent results, costs, capacity, a risk mandate). Requiring a demonstrated PnL before the data can even be examined would be circular. What the trial records from the first day is the delivery dates, a condition of testability and not a profit requirement.

10	91%	0.16	−2.6 ¢
pre-registered tests run once (T1–T5, T1c, T2c, BS2, B0, AR), each with its verdict	of 828 same-pass tank pairs agree within 1 m on roof depth (Sentinel-2, winter)	hub test correlation, 95% interval −0.32 to 0.57 on 19 weeks: inconclusive	WTI move per 1 Mb US ex-Cushing surprise within six minutes (t −6.2); Cushing not detected

State of the work on 26 October 2026.

- IMPLEMENTED A Python toolkit (EIA files and report archive, trading calendars, statistics, Sentinel-1/-2 access, OpenStreetMap tank registry, ENTSO-E, GIE AGSI+, Open-Meteo, HistData minute loader, roof-shadow model), 33 unit tests on synthetic data, three frozen pre-registrations whose hashes the test scripts verify, single-run guards and trial logs.

- EMPIRICAL FINDING EIA report days are about 10% more volatile for WTI (1.10; 1.11 with the true release dates) and natural gas (1.09), less clearly since 2017. A naive surprise-direction test is not supported. Sentinel-2 roof-shadow depths are partly repeatable per tank; the weekly hub-level test is inconclusive. At the print, US ex-Cushing, gasoline and distillate surprises move WTI within six minutes EXPLORATORY; the Cushing coefficient is not detected on the outright, which is not evidence of no effect. A price-only proxy of branch A (a partial exploratory reproduction of the relation studied by Alturki and Kurov) shows a weak positive post-API association in 2013–2019 (r 0.15); persistence is not established in 2019–2023 (r −0.03, 95% interval −0.18 to 0.13) EXPLORATORY.
- PROPOSED The nowcast ladder B1, the price-information test B2, the net-value test B3, a vendor-data trial protocol with a prepared sample request, and a six-month plan with three gates whose outcomes are promising, inconclusive or stop.

Scientific reading so far. External evidence (Section 4.1) shows that inventory information moves prices and that, under the authors' cloud-cover identification, private satellite estimates changed what the official figure told the market; whether a given source adds net value is what the trial measures. The free 10 m optical route measures floating-roof depth partly repeatably, but the weekly hub test is inconclusive and its causes are not identified: cadence (one orbit, clouds, sun angle) is a documented constraint of the current protocol, not shown to be the cause. The outright WTI price moves on inventory news within minutes; the Cushing coefficient is not detected, and its interval contains the effect of the other categories. The open questions are what reaches us before the API and EIA releases, whether the CL1–CL2 spread responds differently, and whether a professional source adds information usable at our action time after costs. Each is tested in the trial before any subscription (Sections 5–9).

Statistical reality. A weekly event strategy makes about 52 decisions a year. The standard error of a forecast–price correlation measured on n decisions is about $1/\sqrt{n}$: telling $\rho = 0.06$ from zero at two standard errors needs about 1,100 decisions, two decades of reports. Six months of new weekly observations are generally not enough to identify so weak an effect; a longer reserved history or a larger effect changes this assessment. The gates therefore rest on higher-information intermediate evidence (measurement error, nowcast error on first-published values, price transmission) and on mechanism; the held-back window is a consistency check, not a proof.

2 Research integrity conventions

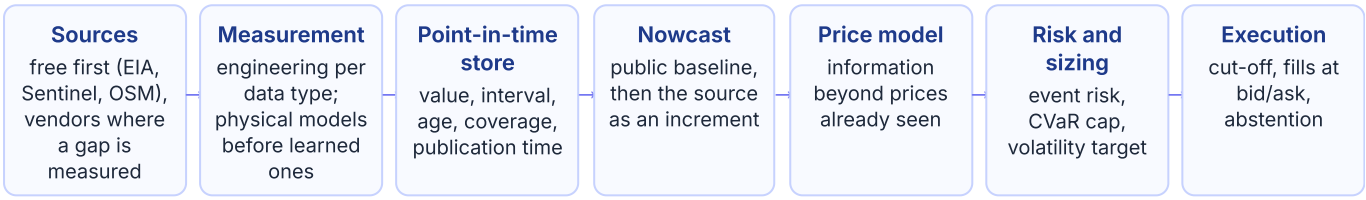
RULE	IMPLEMENTATION IN THE REPOSITORY
Pre-registration	`prereg/tests_v1.json` (T1–T5), SHA-256 `66bec92f8383607c1777e3d69c23aa8abd25efa1f90bead18fc9f22615f1cbbe`; `prereg/tests_v2.json` (T1c, T2c, BS0–BS2, B0), SHA-256 `acfe66efb2878adf811e5b58aea99ff14048199728a9d76cc1045c67da67d84f`, frozen 2026-10-25 01:21 (+02:00) and recorded in `results/v2/prereg_v2.sha256`; `prereg/tests_v3.json` (AR, AH, AC), SHA-256 `d0b8d4c6b9604bd15d0f57037208d5208b4e6747c449c78bf62e76555a824ec9`, frozen 2026-10-25 09:38 (+01:00) and recorded in `results/v3/prereg_v3.sha256`. Scripts stop if the hash differs.
Single run, trial log	Each test ran once. The v2 scripts for T1c, T2c and B0 refuse to overwrite an existing result. Every v2 run appends a line (id, timestamp, script, hash, verdict, seconds) to `results/v2/trial_log_v2.csv`: ids BS1, BS0+BS1, BS2, FIG, T1c, T2c, B0, B0-AM1, B0-AM1-ALLYEARS; the v3 run is in `results/v3/trial_log_v3.csv` (AR).
Deviations before statistics	Implementation choices left open by the pre-registration are dated and written before the statistic they affect: `results/deviations.md`, `results/v2/deviations_v2.md` (13 items for B-S, 9 for T1c/T2c/B0), `results/v3/deviations_v3.md` (8 items for AR, AH, AC).
Amendments	Two, both written before any outcome of the test concerned. AM-B0-1: the B0 validity gate failed; the blocked first run stays in `results/v2/B0/`, the amended run is `results/v2/B0_amended/` and is reported as exploratory. AM-A-1: the CL halt began at 17:15 before 21 September 2015, so the structural clock check of A counts bars from 17:15. One post hoc sensitivity, not pre-registered: B0 with all years (`results/v2/B0_allyears/`).
Verdicts	Three outcomes, fixed in advance: supported / not supported / inconclusive (or signal / no signal / inconclusive). Inconclusive is never a pass.
Point-in-time	Availability timestamps, not event timestamps, define what is known. EIA values are first publications from the report archive; satellite scenes count from their publication time.

RULE	IMPLEMENTATION IN THE REPOSITORY
Secrets and licences	API keys are read from the environment and never printed or committed. Licensed data (HistData minute files) and image caches are not redistributed; scripts rebuild caches from the sources.

3 System architecture, common to the six ideas

The scaffold is IMPLEMENTED for data access, point-in-time storage, statistics and validation; the regime, fusion and risk layers are PROPOSED for I1 and specified below.

Interface contract. Every layer passes, at a timestamp, a value, an uncertainty interval, an age and a coverage flag. A week without an admissible measurement falls back to the public model; uncertainty is propagated, not discarded.
Engineering per data type: optical and radar images use physical forward models first (Section 5) and learned models only when labels exist; text uses embeddings with explicit look-ahead controls (filing dates, anonymised identifiers); flows use sequence models; tabular series use linear models or trees.



Regime layer: observability (clouds, revisits), curve state, volatility regime, calendar events: it weights sources and sizes risk; it is not an alpha.

Validation, constant: pre-registration, walk-forward with embargo, placebos, trial accounting, held-back window kept by Borel.

IDEA	STATUS	MECHANISM AND EVIDENCE IN THIS REPOSITORY
I1 Energy inventories	Prototyped	Physical stock measurement before the EIA report: a free version is built and tested; its final systemic version is the object of the plan (this document).
I2 European power and gas	Prepared	Weather balance moves prices and storage draws. T4: −15.9 EUR/MWh per +10 points of wind and solar share (t = −16.4; `results/T4.json`); T5: +645 GWh/d of EU net withdrawal per heating degree day (t = 37.5; `results/T5.json`). Mechanisms, not signals: forecasts are as published, after the auction.
I3 Equity revenue before earnings	Next	Transaction or geolocation panels nowcast revenue; needs a panel with point-in-time history.
I4 Text, I5 crops, I6 event volatility	Reserve	Same discipline; PnL of I4 not quantified (effective number of bets unknown).

4 I1 specification

4.1 Timeline and hypotheses

EIA stocks are as of Friday 07:00 ET (7 a.m. Eastern Time in the EIA-800, EIA-803 and EIA-804 survey instructions; the frozen pilot protocol used 07:00 Central, and recomputing BS0 and BS2 with the Eastern reference changes no selected scene and no result: `results/v3/et_reference_sensitivity.json`); the industry (API) estimate is published Tuesday 16:30 ET; the EIA Weekly Petroleum Status Report is published Wednesday 10:30 ET, shifted in holiday weeks. Release dates come from the EIA report archive (796 releases since 2011, `audit/eia_release_registry.csv`): the rule calendar used by v1 misdated 14 of 660 weeks in 2011–2024 (`audit/calendar_audit.md`).

EXTERNAL EVIDENCE (NOT OUR RESULTS)	WHAT IT SHOWS	WHAT IT DOES NOT SHOW
Mukherjee, Panayotov & Shon, *JFE* 141(1), 2021. Figures from the open working paper (HKUST IEMS 2019-68, October 2019); the tables of the final version were not compared.	261 weeks in 2014–2018, ten oil hubs. After a positive US stock surprise of one standard deviation, the WTI return from 10:30 to 11:00 is −0.51% in cloudy weeks and −0.05% in clear weeks (difference −0.46, significant at 1%); in 2007–2011 the contrast is not significant (+0.12). Clouds degrade commercial satellite estimates (their Table 4), which is what identifies the effect. Their surprise is the percentage stock change minus the mean of the previous four weeks, not our AR(2) proxy.	A net backtest; Cushing alone; a causal reading independent of the cloud-cover identification.
Alturki & Kurov, *J. Futures Markets* 42(1), 2022.	Out-of-sample predictability of the pre-EIA return by the move after the API release (their Table 2A and Figure 4), January 2011 to August 2019. For a second predictor, API minus the Bloomberg consensus (their Table 8), annualised Sharpe ratios of 0.67 on 385 observations and 2.10 on 77 extreme observations. Table references and values were confirmed in the full text by the independent review of 25 October 2026.	Persistence after the sample; today's costs. The price-only relation tested here (Section 6.5) is a partial exploratory reproduction, not their contracts and data, their full recursive predictive exercise or the strategy of their Table 8. The Sharpe ratios are the authors' historical results.

Thesis B, central. Let Y_t be the first-published weekly change of Cushing stocks, X_t the public information at a decision time τ_t before the release (prices, curve, lagged stocks, season, the API figure if already out), and Z_t a physical measurement available at τ_t. B holds if $\mathrm{E}[\Delta P_t \mid X_t, Z_t]$ differs from $\mathrm{E}[\Delta P_t \mid X_t]$ by more than trading costs, for a tradable price change ΔP_t. It is tested as a chain: **B-S** (does an admissible Z exist, with known error, coverage and age?) → **B1** (does Z improve the forecast of Y beyond a public model?) → **B2** (does that improvement carry price information?) → **B3** (does it survive costs, capacity and the data cost?). A pass at one step proves nothing at the next.

Branch A, complementary. A tests a published price effect: the WTI return in the 30 minutes after the API release against the return from 60 minutes to 1 minute before the EIA release. It is capped in effort and reported separately (Section 6.5). It shares infrastructure with B (events, prices, execution) but a result of A is never counted as evidence for B.

Mechanism and counterparty. Participants who react to the print and hedgers without an informational motive. Two forces work against B: storage arbitrage makes stocks respond to prices (a better stock forecast is not automatically a price forecast), and some satellite information may already reach prices before announcements; vendors already sell measurements. Competition is a question of execution and incremental value, and is measured in the trial: the bar is information beyond prices, not a better picture.

4.2 Targets and surprise proxy

The target is the change of Cushing stocks as first published in the week's report (current minus prior level printed in the same release; `results/v2/eia_first_release.csv`). Cushing first publications equal the latest values in all 795 archived weeks; US commercial crude differs in 272 weeks of 2011–2016 (`audit/vintage_audit.md`). The surprise proxy is $u_t = Y_t - m_0(X_t)$, with m_0 a recursive AR(2) with intercept on first-published changes since 2011-07-22 (at least 104 rows). It is a proxy, not the market consensus. US crude excluding Cushing is US commercial crude minus Cushing in both terms.

4.3 Data contracts

SOURCE	CONTENT	AVAILABILITY RULE AND VINTAGE	LICENCE
EIA report archive	Release dates; first-published weekly stocks (Cushing, US commercial, gasoline, distillate)	Archive folder date; data-only re-issues dropped; earliest release per week	Public domain
EIA NYMEX series	Daily contracts 1–2 (WTI, Henry Hub)	Official settles; series stop on 2024-04-05	Public domain

SOURCE	CONTENT	AVAILABILITY RULE AND VINTAGE	LICENCE
Sentinel-2 L2A	10 m optical, Element 84 Earth Search STAC	Counts from publication: median 6.0 h after sensing, p90 8.6 h, max 12.8 h (149 products in 2026, Copernicus Data Space catalogue)	Free, full and open
Sentinel-1 RTC	Radar amplitude, Microsoft Planetary Computer	At Cushing ascending only: about 55 passes a year in 2019–21, 28–31 since 2022	Free, full and open
OpenStreetMap	Tank outlines	Volunteered; incomplete; roof type not tagged	ODbL
HistData	WTI minute bid bars	Clock converted by period and checked against the CME daily halt (17:15 before 21 September 2015); no file for 2024-01 to 2026-05	Free download, not redistributed
To obtain at Borel	CL1, CL2 and spread quotes with sizes; API and EIA timestamps; the vendor sample (Section 8); forecast runs as issued	Timestamped at receipt; revisions kept apart	Under Borel licences

4.4 Instruments and position generation PROPOSED

- **Instruments.** CL outright and the CL1–CL2 calendar spread, in USD per barrel (the traded unit; equal percentage moves of two maturities are not a zero spread move). Products and options only after B2.
- **Decision time.** Set by the delivery of the measurement (own pipeline or vendor), before the API release when possible; one primary cut-off chosen from coverage and availability, without looking at returns. A post-API variant is a separate experiment.
- **Signal and abstention.** Direction = sign of the forecast price change of the instrument; no position if the forecast edge is below the estimated cost plus a margin fixed on development data. One decision per data week: about 52 a year, fewer without a usable measurement. The outright and the spread are two expressions of one signal, not independent bets.
- **Holding.** From the cut-off to six minutes after the print (primary), 30 minutes after (one pre-declared secondary). Branch A has its own window: entry 60 minutes before the EIA release, exit 1 minute before (5 minutes before as a sensitivity).

4.5 Physical availability: what a total stock hides PROPOSED

Facts. EIA (*Today in Energy*, 16 July 2026): storage facilities need a minimum volume in tanks and pipes to operate (tank bottoms), which varies between facilities, so not every barrel in the statistics is accessible; Cushing stocks were below 20 million barrels from the week ending 19 June to the week ending 10 July 2026, and the Brent minus WTI spot differential turned negative on some days; this is EIA's observation and interpretation, not an identified return. NYMEX Submission 26-389 (14 September 2026), Exhibit G (supplemental market information), estimates deliverable supply separately from EIA stocks: 40% of inventory is discounted for qualifying streams (PDF page 11), then deliverable supply is reduced by 10% for volumes segregated for end-user refiners (page 13). These are conservative regulatory assumptions, not proportions observed each week, and not coefficients for a signal.

Hypothesis for investigation. The distribution of stock across facilities, its mobilisation and its trajectory may carry financial information that the weekly total does not. Three objects are kept apart:

OBJECT	INFORMATION SOUGHT	CANDIDATE EXPRESSION
Change in total stock	Revision of the local balance, possibly informative about the US balance	WTI directional; the spread as comparator
Trajectory between publications	Acceleration of outflows, reversal, new flow constraints	Return from the delivery of the data; horizon defined with the product
Local availability	Concentration of low levels, access to networks, approximate composition	Nearby maturities, calendar spread; a local basis if an exploitable market is accessible

The first trial uses one principal signal that the product can actually build (the change in total stock against the public nowcast) and at most one physical alternative, motivated before any result (an availability proxy). A by-operator series

does not identify quality, ownership, contractual commitments or the availability of each barrel: an indicator of tension is a proxy whose usefulness is to be evaluated. No preference for spreads is assumed: it depends on the response across maturities and on costs, and a spot Brent–WTI observation does not prove the return of a futures spread.

What the repository holds, and what the availability indicator lacks. Registry (`results/tank_registry.csv` , OpenStreetMap): id, position, diameter and area for 333 tanks; a content tag on 2, an operator tag on none, a roof tag on none; no height, capacity or product. Per tank and date (`results/v2/BS1/fits.csv.gz`): status, depth with interval, lit and shadow levels, clear flag, sun angles; a median 62% of dates carry an estimate per tank (10th to 90th percentile 53% to 66%), and 4 of 234 tanks are full-or-fixed on at least half of their dates. Per week (`results/v2/BS2/weekly_table.csv`): a scene is selected in 159 of 411 weeks and a volume change exists in 62, with a median valid roof-area share of 0.81 and a median gap of 28 h (90th percentile 43 h) between the scene and the Friday stock reference (measured against the frozen protocol's 07:00 CT; with 07:00 ET each gap shifts by one hour). The BS0 indicator is a scene-level yes/no (a usable pass within two days of Friday before the cut-off). It does not say which tanks are covered each week, nor their operator, roof type, capacity, content or membership in the EIA perimeter, nor how the OpenStreetMap registry changed over time: these need by-tank vendor data or a verified registry.

5 B-S: physical measurement from free Sentinel-2 images

Question. Can floating-roof depths at Cushing be measured from 10 m images, repeatably and in time, before any return or EIA error is looked at? **Scope.** The OpenStreetMap registry holds 333 tanks in the 6.8 × 6.0 km window; the 234 with a diameter of 40 m or more are modelled. Roof types are not tagged: a fixed roof shows no internal shadow and ends as *full_or_fixed*.

5.1 Forward model IMPLEMENTED src/altdata/tankshadow.py

When a floating roof sits a depth d below the rim, the wall on the sun side casts a crescent on it of width

$$s = d / \tan\theta, \qquad \theta = \text{sun elevation}.$$

For each interior pixel p (at least 90% of its 100 sub-points inside the disk), the shadowed share $f_p(d)$ is computed on a 1 m sub-grid for $d \in \{0, 0.5, \ldots, 20\}$ m, and the red band is modelled as

$$x_p = a\,(1 - f_p(d)) + b\,f_p(d) + \varepsilon_p, \qquad b \le a,$$

with lit-roof level a and shadow level b re-estimated by least squares for each tank and date, which absorbs some radiometric variation within the model; registration, mixed pixels and scene-dependent errors remain. The estimate $\hat{d}$ minimises the residual sum of squares over the grid; the nominal 68% interval is $\{d: \mathrm{RSS}(d) - \mathrm{RSS}_{\min} \le \hat{\sigma}^2\}$. One translation per scene registers the registry to the image (correlation with a synthetic image of all 333 disks, ±2 pixels, parabolic refinement). Abstention, in order: *few_pixels* (< 8 interior pixels), *cloudy* (cloud or shadow classes within the disk plus 20 m), *no_contrast* ($a - b < 3\hat{\sigma}$), then kept with a flag: *full_or_fixed* ($\hat{d} = 0$) and *wide_interval* (> 6 m). The hub proxy is the volume change under the roofs of the tanks measured at both dates, $\Delta V = -\Sigma_i A_i(d_{i,F} - d_{i,F-1}) \times 6.2898$ bbl/m^3, with A_i the roof area.

Orientation check before any outcome (`results/v2/BS1/geometry_check.json`): on three winter scenes, the median RSS ratio to a constant fit is 0.153 with the physical sun direction and 0.998 with the mirrored one (701 fits).

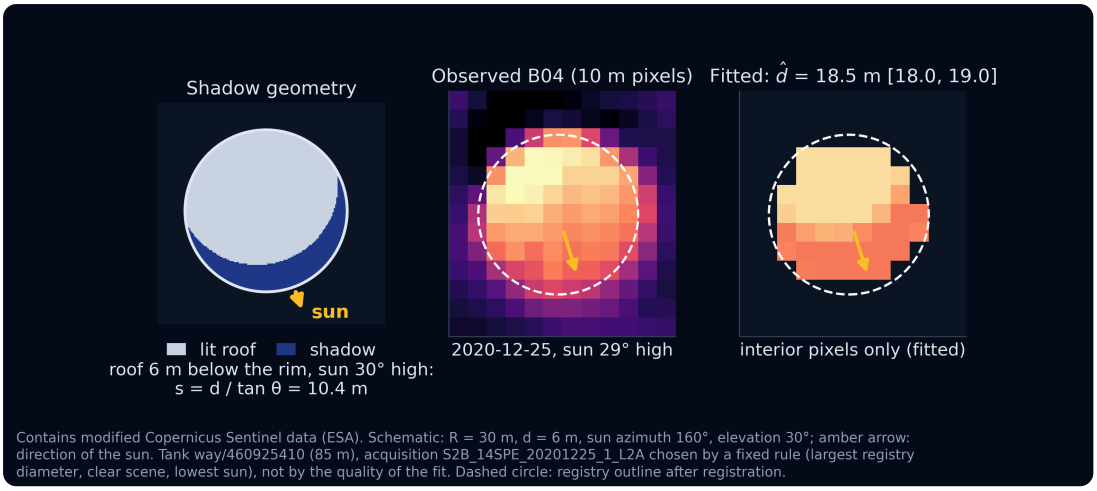


Figure 1. Geometry, observed pixels of the largest registry tank on its lowest-sun clear scene (chosen by a fixed rule, not by fit quality), and the fitted model. `figures/BS1_shadow_model.png` . Contains modified Copernicus Sentinel data (ESA).

5.2 Results EMPIRICAL FINDING descriptive tests BS0 and BS1; results/v2/BS0, results/v2/BS1

QUANTITY	VALUE	FILE
Sentinel-2 acquisitions	603 (2018-12 to 2026-10), all on relative orbit R112; 441 with scene cloud ≤ 90% processed	`BS0/catalog_summary.json`
Usable pass within ±2 days of Friday, before the Tuesday cut-off	Sentinel-2: 39% of 410 weeks; 14% with sun ≤ 45°; Sentinel-1: 44% (24 h latency assumed)	`BS0/availability_summary.json`
Tank-dates with an estimate	60% of 103,194; cloudy 33%; no contrast 7%	`BS1/coverage_summary.json`
Repeatability, same pass	828 pairs imaged 27 s apart by two satellites (4 winter dates, sun 31–36°): 91% within 1 m, 78% within 0.5 m	`BS1/stability.csv`
Registration	Median |shift| 1.2 m (x), 2.4 m (y); 7% of scenes flagged at the search border	`BS1/coverage_summary.json`

SUN ELEVATION	25–35°	35–45°	45–55°	55–65°	> 65°
Simulated RMSE of $\hat{d}$ (m) ILLUSTRATIVE	2.28	3.20	4.48	6.07	7.21
Coverage of the nominal 68% interval	0.33	0.38	0.44	0.52	0.57

Simulation (`BS1/identifiability.csv`): 30 catalogue scenes × 40 registry tanks × depths 0–14 m per sun bin, centre offsets up to 5 m, noise and contrast drawn from the real fits. Contrast is taken from identified fits only, so the simulation flatters the method.

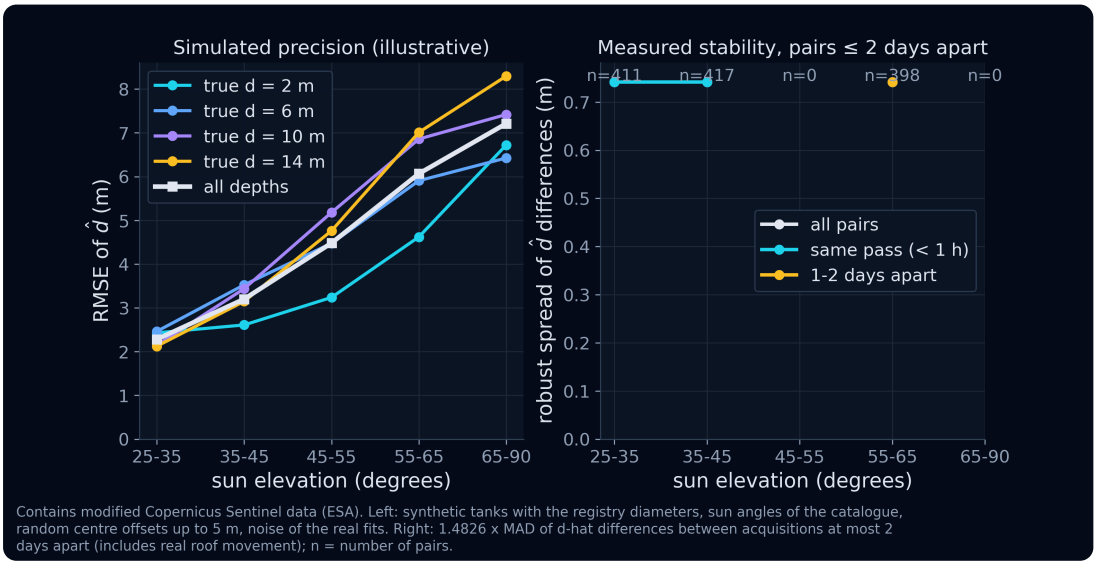


Figure 2. Simulated precision by sun elevation (left) and measured stability of pairs at most two days apart (right). `figures/BS1_identifiability.png` .

5.3 Hub-level test BS2 EMPIRICAL FINDING results/v2/BS2/bs2.json

Rule fixed in advance: signal if the lower 95% bound of the correlation between ΔV and the first-published EIA change is above 0; no signal if the upper bound is below 0.20; otherwise inconclusive. Primary sample: both scenes with sun ≤ 45° and measured tanks covering at least half of the roof area.

SAMPLE	WEEKS	R	95% INTERVAL (FISHER)	BLOCK BOOTSTRAP	VERDICT
Primary, low sun	19	0.16	−0.32 to 0.57	−0.32 to 0.67	**Inconclusive**
All seasons (secondary)	57	0.13	−0.14 to 0.38	−0.11 to 0.37	**Reported**

Of 411 weeks, 159 have a selected scene and 62 a volume change; the nowcast secondary test was not evaluable (fewer than 20 earlier valid weeks).

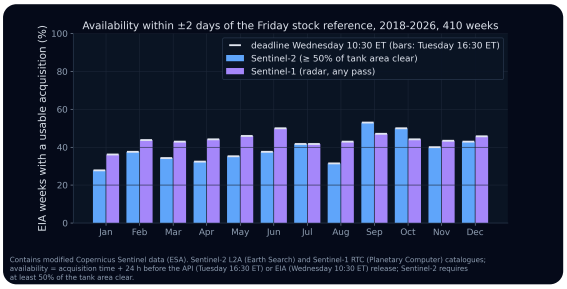


Figure 3. Weeks with a usable pass, by month. `figures/BS0_availability.png`.

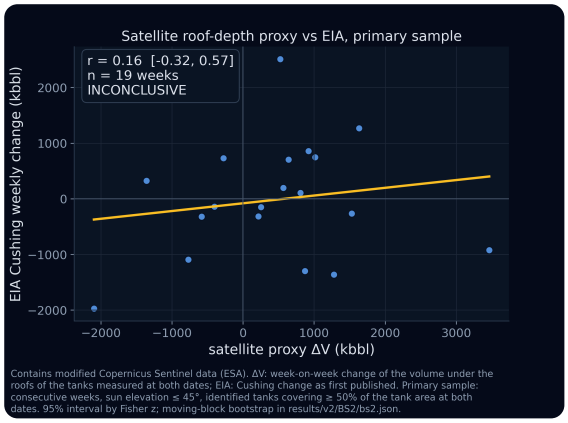


Figure 4. Hub proxy against the first-published change, primary sample. `figures/BS2_proxy_vs_eia.png`.

5.4 What is and is not established

- **Established:** a physically oriented, partly repeatable roof-depth measurement per tank on low-sun scenes, from free data. It is an informative measurement experiment.
- **Not established:** accuracy against operator gauges (none available: repeatability is not accuracy), a weekly hub variable that tracks EIA, and the cause of the weak hub correlation.
- **Error model:** the nominal intervals are too narrow (33–57% coverage for 68%), so $\hat{\sigma}$ understates the error (registration, point-spread function, roof structure). The day-to-day robust spread, 0.74 m, equals 1.4826 times the 0.5 m grid step: the grid, not the method, limits what the stability table can resolve.
- **Hub test:** independent tank noise averages out over 234 tanks; scene-wide errors do not. Candidate causes of the weak correlation, none of them identified: few Friday-matched passes, up to two days between scene and stock date, scene calibration, tanks outside the EIA scope, and a weekly hub change of only about 1 Mb.
- **Power:** detecting $r = 0.3$ with 80% power at the 5% level needs about 85 weeks; at 14% usable low-sun weeks, that is more than a decade under the current protocol. Cadence is a documented constraint of this protocol (two passes near consecutive Fridays), not shown to be the cause of the weak correlation, and not a limit of Sentinel in general: asynchronous observations remain usable with interpolation or a state-space model, to be tested.

5.5 Next steps for B-S PROPOSED

These steps follow the professional-data trial (Section 8): they are taken up where the trial identifies a limit they can address, and meanwhile run only at low cost on free data.

1. Scene calibration with references inside each scene (fixed roofs, ground), and an empirical error model fitted on repeat pairs instead of $\hat{\sigma}$.
2. Friday anchoring: interpolate depths to the stock date with an interval that grows with the gap.
3. Sentinel-1 phase data (SLC): roof-height methods from coherent scatterers exist for TerraSAR-X (Villamil Lopez and Stilla, 2021); they are not validated on Sentinel-1 and would be tested as such.
4. A vendor trial evaluated against the pilot on overlapping weeks (Section 8), and ground truth on a sample: operator gauges or very-high-resolution tasking.

6 Price experiments: B0 to B3, and branch A

6.1 B0: transmission at the print EMPIRICAL FINDING EXPLORATORY

Design (`scripts/b0_transmission.py`, deviations items 3–8). Events: Wednesday 10:30 releases dated by archive folder, 2013-08 to 2023-12 and 2026-06 to 2026-10 (562 candidates; excluded: 81 holiday-shifted, 2 data-only days, 15 without minute data, 3 without a surprise). Outcome: ΔP = close of the 10:35 bar minus close of the 10:29 bar, i.e. the WTI bid change from 10:30:00 to 10:36:00 ET, in USD per barrel. Regressors: the four first-published surprises (Cushing, US ex-Cushing, gasoline, distillate), each the error of its recursive AR(2), in million barrels. OLS with HC1 errors. Rule: transmission detected if the 95% interval of the Cushing coefficient lies below zero.

Validity gate and amendment. The pre-registered clock check (10:30 the largest one-minute move of 10:00–10:59 on at least 70% of days) failed: 205 of 464 days, 44%, although the average profile peaks at 10:30 (24.1 cents against 3–4 cents in the 15 minutes before). An argmax rule fails whenever volatility stays high for several minutes, so it did not test the clock. Before any outcome, AM-B0-1 replaced it by a year-wise gate (the mean profile must peak at 10:30 at more than three times its 10:15–10:29 mean). All years 2013–2023 pass; 2026 fails (10.9 cents against 5.5) and is excluded from the amended run. B0 is therefore exploratory.

Critique and sensitivity. The gate excluded 2026 because its 14 events had a weak reaction at 10:30. A weak reaction is not evidence of a wrong clock, and independent evidence supports the file clock in every year: the CME halt (bars from 17:15 to 17:59 are absent on API days in all twelve years, 0.008 of hour-16 bars at most; the bars from 17:00 to 17:14 vanish after 21 September 2015, as the halt moved from 17:15 to 17:00), the Sunday opening at 18:00 New York on 532 of 547 Sundays, the EURUSD releases at 08:30 peaking at the stamped minute in 2016, 2019, 2022 and 2026, and the API release at 16:30 (mean absolute change of the 16:30 minute 4.7 to 20.8 cents against 1.1 to 3.1 over 16:15–16:29, in all twelve years; `results/v3/a_price_proxy.json`, AC). A post hoc sensitivity keeping every year (461 events, `results/v2/B0_allyears/`): Cushing −3.16 cents per Mb, interval −6.52 to +0.20 (t −1.84), US ex-Cushing −2.52 (t −6.18), R^2 0.21. The conclusion is unchanged.

SURPRISE (1 MB)	CENTS PER BARREL	95% INTERVAL	T	SURPRISE SD (MB)
Cushing	−3.07	−6.56 to +0.43	−1.72	1.16
US crude ex-Cushing	−2.62	−3.44 to −1.80	−6.25	4.89
Gasoline	−3.58	−5.00 to −2.17	−4.96	2.77
Distillate	−3.10	−4.53 to −1.66	−4.24	2.45

447 releases, 2013–2023; R^2 0.21; sd of ΔP 39.1 cents. Secondary horizon (to 11:01, 436 releases): Cushing −1.42 (t −0.65), ex-Cushing −1.82 (t −3.07), R^2 0.07. `results/v2/B0_amended/b0.json`; verdict NOT DETECTED.

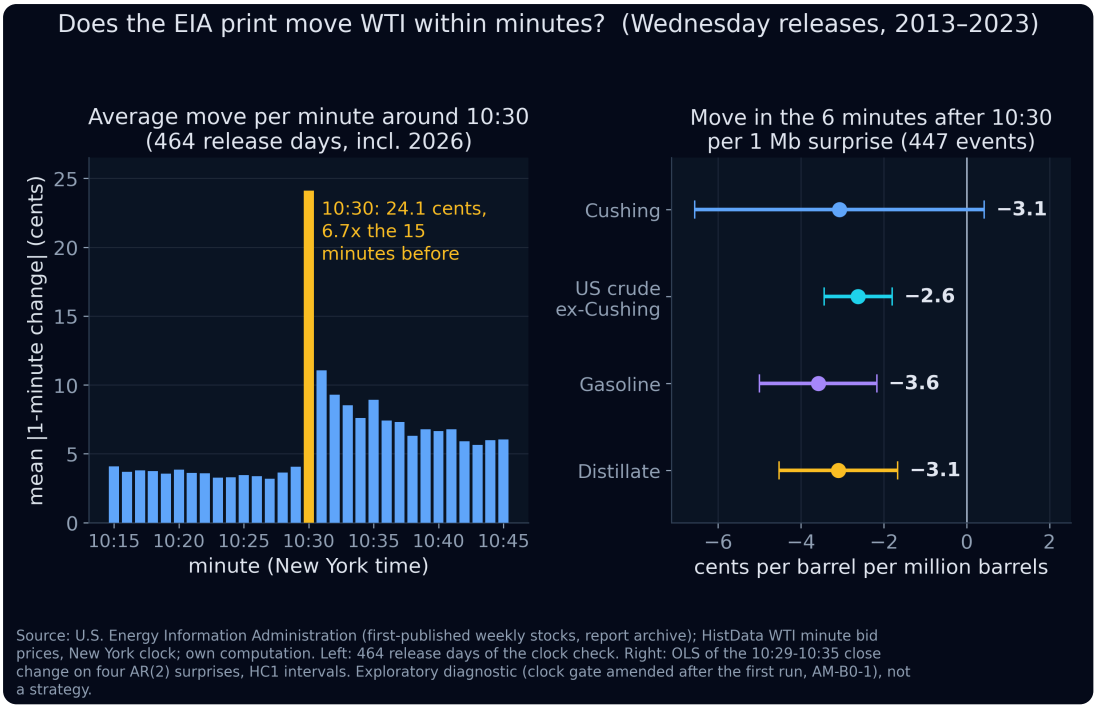


Figure 5. Left: mean absolute one-minute change on the 464 release days of the clock check. Right: coefficients with 95% intervals. `figures/B0_transmission.png` .

Reading. Inventory news moves WTI within minutes, with similar per-Mb effects across categories. A reaction in a window does not show full incorporation, and a non-detection is not an absence of effect: the Cushing interval (−6.6 to +0.4) contains zero and also the −2.6 estimated for the other categories. Cushing surprises are small (sd 1.16 Mb against 4.89 for the rest of US crude), which limits power. **Limits:** the surprise is a proxy, not the consensus (the published work uses the percentage change minus a four-week mean, another proxy); bid-only CFD prices; outright only, no spread; no data for 2024-01 to 2026-05. **Scope:** a regression on the realised surprise is an ex-post diagnostic of transmission; a position built on *u* would be an oracle, not a backtest.

6.2 B1: incremental nowcast value PROPOSED

MODEL	CONTENT	STATUS
B1a	AR(2) on the weekly change, expanding window	FINDING out-of-sample R^2 0.307 on 521 forecasts (final series), 0.324 on 517 first-published values; RMSE 1.20 Mb against 1.44 for a constant mean; sign hit 0.67; ±1.2 Mb holds 74% of errors (`results/v2/baseline_cushing/metrics.json`)
B1b	Public model: season, lagged stocks, curve, prices already seen, API if known at the cut-off	To build; the benchmark that matters
B1c	B1b plus *Z*, its age, coverage and uncertainty	Requires an admissible *Z* (B-S)

Walk-forward with an embargo; normalisation and calibration on the past only; extraction variants counted as trials. Metrics: RMSE, MAE, bias, interval calibration and the paired loss difference with a moving-block bootstrap (blocks of four weeks). **Promising** if the 95% interval of the loss difference favours B1c over B1b on the evaluation window and the gain holds at realistic cut-offs and in separate periods; **inconclusive** if the interval cannot separate a useful gain from none (one bounded extension is justified only if it can change the decision at a reasonable cost); **stop that use** if the data are unfit for the test, a useful gain is excluded, or the extra information would cost disproportionately: a budget decision, not proof that no edge exists. Weeks without *Z* stay in the universe: both common-coverage and operational (fallback) performance are reported. A noisy source can still help the residual of a precise benchmark; its raw error need not beat 1.20 Mb.

Target-adapted. B1 is a diagnostic of fundamental value for the target the source is built to measure. For a stock-change measurement the target is the first-published Cushing change and the comparison is B1c against B1b. For a source built to measure another physical constraint (for example local availability) the target and the loss are declared

before estimation: the mean RMSE of the EIA figure must not eliminate it. One principal signal and at most one physical alternative, both fixed before results; any change made after seeing results is exploratory.

6.3 B2: information beyond prices PROPOSED

Compare $g_0(X)$ and $g_1(X, a)$ on the price change of each instrument from the cut-off to six minutes after the print, where a is the B1 correction produced out of fold (never fitted on the period it predicts), and X includes the curve, past returns and, after the API, the post-API move used by A. The test statistic is ρ, the out-of-sample correlation between the forecast and the price change, and value is judged incrementally: g_1 is compared with g_0 on the same decisions by a paired difference with a block bootstrap, because a good total correlation does not show that the source adds anything. For constant-size bets on the sign of the forecast (the Gaussian illustration of Section 9) the cost threshold is

$$\rho^* = k / \sqrt{(2/\pi)} \approx 0.025 \text{ for } k = 0.02 \text{ (CL)}.$$

Promising if the paired incremental gain of g_1 over g_0 has a lower 95% bound above zero and the augmented forecast's ρ is above ρ^*; **inconclusive** if the interval cannot separate a useful gain from none; **stop that use** if a useful incremental gain is excluded (a budget decision, not proof that no edge exists). ρ^* belongs to the sign-rule illustration and is not a universal criterion for every policy; B3, the incremental net PnL against the same strategy without the source, remains the economic arbiter. CL and CL1–CL2 are measured separately; the instrument is chosen on development data, then frozen. With five years of weekly decisions ($n \approx 260$) the standard error of ρ is about 0.06, so only large effects can be confirmed; small ones will remain inconclusive, and that will be reported as such.

6.4 B3: net value against the same strategy without the source PROPOSED

Policy: one unit in the direction of the forecast price change, abstention below cost plus a margin fixed on development data. Counterfactual: same calendar, risk rules and costs with the public model only; the value of the source is the PnL difference, not the total PnL. Fills at bid and ask with sizes, fees, latency, weeks without measurement, size limits. Reported: mean per eligible event and per trade with intervals, drawdown, tail risk, concentration by year, abstention rate, break-even cost. Capacity from depth and impact, then the maximum data budget. Fifty-two announcements do not become 104 independent bets by adding a second instrument.

Defined before estimation: the principal instrument and its comparator, a realistic action time, the horizon, the reference without the source, the construction sample and the reserved period. **Outputs:** relevant measurement error, contribution to the signal, signed mean move reachable, uncertainty, incremental gain after execution and the data cost. Information useful only before delivery, or absorbed before our action time, directs a technical decision or the end of that specific use. A source failure is not extended to all physical information, and trials are not multiplied until a favourable statistic appears.

6.5 Branch A: a price-only proxy EMPIRICAL FINDING EXPLORATORY prereg/tests_v3.json; results/v3/a_price_proxy.json

Design (`scripts/a_price_proxy.py` , `results/v3/deviations_v3.md`). Events as in B0 (562 candidates; 81 holiday-shifted, 2 data-only days, 51 with a missing bar; 428 included). r_{api} = 100 ln[P(17:00) / P(16:30)] on the day before the release; r_{pre} = 100 ln[P(10:29) / P(09:30)], with $P(t)$ the close of the bar starting one minute earlier. Rule fixed in advance on the overlap block (2013-08 to 2019-08, the part of the authors' sample that our data cover: the same weeks with another instrument and price source, so a proxy check and not an independent replication): signal if the lower 95% Fisher bound of r is above 0; no signal if the upper bound is below 0.10; otherwise inconclusive. The post block (2019-09 to 2023-12) and 2026 are descriptive. No event and no year is dropped for the size of a move or a weak reaction; the clock is validated structurally (Section 6.1) and by amendment AM-A-1.

BLOCK	N	R	95% INTERVAL	R^2	OLS SLOPE (HC1 INTERVAL)	SPEARMAN	SIGN RULE, GROSS CENTS PER EVENT (T)
Overlap 2013–19	251	0.152	0.028 to 0.270	2.3%	0.109 (0.001 to 0.216)	0.19	+3.81 (2.64)
Post 2019–23	163	−0.026	−0.179 to 0.128	0.07%	−0.028 (−0.362 to 0.305)	−0.04	−3.86 (−1.03)
2026	14	0.699	0.268 to 0.897	49%	2.877 (1.528 to 4.227)	0.66	+29.9 (1.78)

Verdict AR (overlap): SIGNAL, as an exploratory proxy. In the weeks shared with the authors' sample the post-API move predicts the pre-EIA move with a small positive correlation (R^2 2%); the lower bound of r is 0.03 and that of the

HC1 slope 0.001, so the result is weak. After the published sample persistence is not established: the estimate is close to zero (r −0.03, interval −0.18 to 0.13), which does not exclude a correlation of 0.10, and the sign rule's gross gain is negative on that descriptive block. The 14 events of 2026 show a large correlation but are too few to read and fall in an unusual volatility regime. The sign rule's gross mean (3.8 cents per event in the overlap) exceeds an assumed round-trip cost of 1.5 to 2.5 cents, but it is not a net PnL: bid-only CFD bars, no fills, sizes or latency, an assumed cost. In the post block it is −3.9 cents. The published Sharpe ratios concern an API-minus-consensus signal and are not tested here (no timestamped consensus). A result of A is not evidence for B.

Not done, and proposed: the work above is a partial exploratory reproduction, on a price proxy, of the relation studied by Alturki and Kurov (post-API return as predictor of the pre-EIA return: their Table 2A and Figure 4). It does not reproduce their contracts and data, their full recursive predictive exercise or the strategy of their Table 8; that strategy needs a timestamped consensus, which the price relation itself does not. A faithful replication (**A-R**) would transcribe their definitions, windows and tables. **A-P**, tradable persistence, would use CL quotes with a convention fixed in advance (the halt is at 17:00 ET, at 17:15 before 21 September 2015), entry at EIA −60 minutes, exit at −1 (−5 as a sensitivity), aggressive one-lot orders after 1 s latency and a 10 s stress ASSUMPTION, against no position, the recursive mean and a calendar-only rule.

7 Validation, decision rules and statistical limits

- **Sample status.** Development (seen by the researcher), diagnostic (transition periods), held-back (kept by Borel, never inspected). A period already seen is exploratory, whatever its date.
- **Trial accounting.** Every extraction, model and instrument variant is a trial and is logged; selection uses development data only; the selected variant is frozen before the evaluation window is opened.
- **Multiple testing.** Families are declared in advance (e.g. Holm across the two T1 markets); exploratory results are labelled and never used as confirmation.
- **Gates.** Each gate ends promising, inconclusive or stop, by a rule written before the data. Inconclusive is not a pass: it triggers at most one bounded extension of a stated kind, if that can change the decision at a reasonable cost, or a stop of that use. A stop can be an economic decision without a statistical refutation; it is recorded as a budget decision, not as proof that no edge exists, and a negative B1 concerns the intended use and target, not all physical information.

QUANTITY	ARITHMETIC	CONSEQUENCE
Decisions a year (I1)	About 52, fewer without a usable measurement	PnL accrues slowly
Standard error of a measured correlation	$\approx 1/\sqrt{n}$: 0.14 on one year, 0.06 on five	One year cannot tell 0.06 from 0
Decisions to tell ρ = 0.06 from 0 at 2 se	$(2/0.06)^2 \approx 1{,}100$, about 21 years	Early gates rest on B-S, B1 and B0; PnL accrues slowly
Weeks to detect r = 0.3 (80% power, 5%)	$((1.96 + 0.84) / \operatorname{atanh} 0.3)^2 + 3 \approx 85$	Hub test needs many more usable weeks

What would falsify B. No admissible weekly variable with a known error at a usable cadence (B-S); no out-of-sample gain over the public model (B1); a gain in stock forecasts with no price information beyond X (B2); a price effect that does not survive costs, capacity or the data cost (B3). If several physically credible sources fail successively against a solid benchmark, the thesis itself is reduced or abandoned.

8 Professional data: a trial first, a subscription only if earned

PRODUCT	DOCUMENTED BY THE VENDOR	THE TRIAL MUST ESTABLISH
Energy Aspects, TankWatch Cushing	Four reports a week, delivered within 24 h of capture; inventories per tank and per operator, presented as a proxy for the crude mix available for delivery against WTI NYMEX	Delivered history and capture dates, asset identity, levels and changes, limits of the quality proxy. Vendor statement: in constant operation since 2018, with access to extensive historic back-test data (page read on 26 October 2026); a vendor claim, not a guarantee of a Cushing history as delivered.
Wood Mackenzie, Cushing Crude Oil Storage Report	Tank, operator and location level; fixed and floating roofs; infrared cameras and aerial imagery; a methodology refined since 2009 (vendor page, 7 January 2025; the detailed report sits behind a form and was not downloaded)	Coverage that the free pilot lacks; first deliveries and revisions
Kpler, Cushing Drone Inventories (comparator)	Optical and infrared observations three times a week	Realised calendar, coverage, calibration of changes, archive as delivered
Ursa Space (alternative)	Monday delivery; a vendor-reported 0.98 correlation with EIA levels in 2023 (12 February 2024)	Accuracy on changes; level correlations overstate

The first trial concerns the source with the most usable history at an acceptable cost, not necessarily the finest resolution: TankWatch and Wood Mackenzie for economic granularity, Kpler as comparator of frequency and coverage, Ursa if its archive fits better. Raw radar (ICEYE) is not a priority: it reintroduces a measurement project. This is vendor documentation only: prices, rights and the performance for I1 are unknown, and a marketed history can be recomputed, so its age does not guarantee point-in-time availability.

DECISION	EVIDENCE THAT JUSTIFIES IT
Fund an experiment	A documented mechanism (Section 4.1), an uncertainty the experiment can resolve, a bounded cost and scope. Delivery dates are recorded from the first day so that the data are testable; persistence after receipt is measured during the trial, neither presupposed nor required to authorise it.
Subscribe for a strategy	Incremental net value measured on that source's own data, after execution and the data cost.
Commit market risk	Independent results, costs and capacity assessed, and a risk mandate.

Sample request PROPOSED, prepared and not sent, no purchase engaged: (1) a continuous period and the missing weeks, not only successful cases; (2) capture, processing and first-delivery timestamps, and the archive of corrections; (3) stock per tank and operator and the aggregate, with the historical perimeter, roof types and changes of coverage; (4) the estimation method, known limits, uncertainty and available validation references; (5) a comparison with EIA changes that separates measurement dates from publication; (6) trial terms, a quote, and the rights to backtest and to publish aggregate statistics. A short sample can certify the format and the timing; it cannot establish a Sharpe ratio. A longer historical trial would compare the measurement with its financial value. Evaluation code is frozen before the data arrive; no delay in obtaining access is promised.

> **Economics.** Three costs are kept apart: the cost of the trial, a production subscription, and the marginal cost if Borel already holds a licence. Break-even edge per covered event, in dollars per barrel (multiply by 100 for cents): $c_{\min} = C / (q \times 1{,}000 \times \kappa \times 52)$ for an annual cost C, q contracts and a covered share of weeks κ. It is the incremental gain required after variable costs to cover that fixed cost (operating costs are added to C when included), compared with the edge measured on the same horizon, with k set on the holding window (Section 9). The PnL grids of Section 9 do not subtract a subscription or operating costs and are neither an expected-return estimate nor a capacity measure. A larger size amortises a fixed cost and increases risk; it creates no alpha. ILLUSTRATIVE

CENTS PER BARREL PER COVERED EVENT	C = 15,000 USD	C = 30,000 USD	C = 60,000 USD
5 contracts, κ = 0.4	14.4	28.8	57.7
10 contracts, κ = 0.4	7.2	14.4	28.8
20 contracts, κ = 0.4	3.6	7.2	14.4

CENTS PER BARREL PER COVERED EVENT	C = 15,000 USD	C = 30,000 USD	C = 60,000 USD
10 contracts, C = 30,000: κ = 0.2 / 0.6			28.8 / 9.6

9 Execution, costs, risk, capacity and capital

- **Execution simulation.** Fills at bid and ask with displayed sizes, latency 1 s with a 10 s stress, fees per contract; a missing exit quote never deletes an open trade (pre-declared liquidation rule, cost kept) ASSUMPTION.
- **Cost.** k = round-trip cost / sd of the move over the holding window. With a cost of 20 USD per contract (about one tick plus fees; ASSUMPTION) and the sd of the WTI move measured on minute data, 2013–2023 (`results/v3/a_price_proxy.json`, AH): from the API release at 16:30 to 10:36, sd 113 cents (k = 0.018, ρ^* = 0.022); from 09:30 to 10:36, 50 cents (0.040; 0.050); the six minutes after the print, 39 cents (0.051; 0.064). The sd varies by regime (API release to 10:36: 86 cents in 2013–2019, 146 in 2019–2023, 220 in 2026), so k is re-measured on the window and period of the trial and replaced by Borel's costs.
- **Risk.** Event risk: size so that a two-sd move over the holding window costs a fixed share of the risk budget, measured on that window; the six minutes after the print (sd 39 cents) are a floor, not the whole risk. CVaR of the holding window as a position cap; volatility targeting; no position across an uncertain release time.
- **Capacity.** Assumed not binding at a few M$ of annual risk in CL; to be measured on order-book data ASSUMPTION.
- **Capital.** Backtest, then an isolated logged paper run, then 25%, 50% and 100% of target risk. Proposed defaults, to be set with Borel's risk team: halve risk at a drawdown of one annual sd of strategy PnL, stop at 1.5. Capital only under Borel's mandate and approval.

Conditional PnL ILLUSTRATIVE. For constant-size bets on the sign of a forecast (Gaussian approximation), with N independent decisions a year and annual risk budget B:

$$\mathrm{IR} = \sqrt{N}\,(\sqrt{(2/\pi)}\,\rho - k), \qquad \mathrm{PnL} = \mathrm{IR} \times B.$$

For I1 (N = 52, B = 1 M$), by holding window (`results/v3/pnl_grid_windows.csv`, `python scripts/pnl_grid.py --windows`), M$ a year at ρ = 0.03 / 0.06 / 0.10:

HOLDING WINDOW	SD (CENTS)	K	P^*	PNL (M$)
API release 16:30 to 10:36	113	0.018	0.022	0.05 / 0.22 / 0.45
09:30 to 10:36	50	0.040	0.050	−0.11 / 0.06 / 0.29
10:30 to 10:36 (after the print)	39	0.051	0.064	−0.19 / −0.02 / 0.21

The deck's headline figures (0.0 / 0.2 / 0.4) use k = 0.02, the first window (`results/v2/pnl_grid_v2.csv`: 0.03 / 0.20 / 0.43). ρ is the forecast–price correlation measured in B2, not a correlation with the stock surprise. The first grid counted 52 oil and 52 gas reports (N = 104); it is kept in `results/pnl_grid.csv`. These are orders of magnitude conditional on ρ, never results.

10 Results registry

TEST	STATUS	RESULT	FILE (TRIAL ID)
T1	FINDING	Report-day |return| ratio: WTI 1.105 [1.025, 1.189], 743 days, Holm p 0.017; gas 1.088 [1.007, 1.175], p 0.052; placebos 0.963, 0.940; 2017–24: WTI 1.083 [0.972, 1.199], gas 0.996 [0.885, 1.121] — **supported**	`results/T1.json`
T2	FINDING	Return on the standardised surprise: WTI −17.0 bp per sd, t −1.26; gas −8.9, t −0.65 — **not supported**	`results/T2.json`
T3	EXPLORATORY	Cloudy weeks: ratio 1.15 [0.91, 1.47], OLS t 0.88, 430 weeks — **inconclusive**	`results/T3.json`
T4, T5	FINDING	Merit order −15.9 EUR/MWh per 10 points (t −16.4); storage draw +645 GWh/d per HDD (t 37.5) — **supported**	`results/T4.json`, `results/T5.json`

TEST	STATUS	RESULT	FILE (TRIAL ID)
Audit	FINDING	T1–T5 re-run with identical outputs; snapshot equals the folder except two new scripts; calendar 14/660 misdated; Cushing never revised	`audit/repro_T1_T5.md`, `audit/*_audit.md`
T1c	FINDING	T1 with archive dates (12 days moved): WTI 1.108 [1.028, 1.192] — **supported**	`results/v2/T1c.json` (T1c)
T2c	FINDING	WTI −17.9 bp per sd, t −1.32 — **not supported**	`results/v2/T2c.json` (T2c)
BS0, BS1	FINDING	Descriptive: Section 5.2	`results/v2/BS0`, `BS1` (BS1, BS0+BS1)
BS2	FINDING	r 0.16 [−0.32, 0.57], 19 weeks — **inconclusive**	`results/v2/BS2/bs2.json` (BS2)
B0	FINDING	First run blocked by its clock gate (44% < 70%)	`results/v2/B0/` (B0)
B0-AM1	EXPLORATORY	Cushing −3.07 [−6.56, 0.43] — **not detected**; ex-Cushing −2.62 (t −6.25)	`results/v2/B0_amended/b0.json` (B0-AM1)
B0 all years	EXPLORATORY	Post hoc sensitivity, 461 events: Cushing −3.16 [−6.52, 0.20], ex-Cushing −2.52 — **not detected**	`results/v2/B0_allyears/b0.json` (B0-AM1-ALLYEARS)
AR	EXPLORATORY	Price-only proxy of A. Overlap 2013–19, 251 events: *r* 0.152 [0.028, 0.270] — **signal** (weak); 2019–23, 163 events: −0.026 [−0.179, 0.128], persistence not established (the interval does not exclude 0.10); 2026, 14 events: 0.699 (too few)	`results/v3/a_price_proxy.json` (AR)
AH, AC	FINDING	Descriptive: sd of the move by holding window (Section 9); clock evidence by year (Section 6.1)	`results/v3/a_price_proxy.json`

11 Six-month roadmap PROPOSED

WHEN	WORK	GATE (PROMISING / INCONCLUSIVE / STOP)
Month 1	Professional Cushing trial against the pilot (sample, then history); B1 against B1b on the declared target; CL1–CL2 quotes; A capped; I2 in parallel on free data	**Gate 1:** a physical variable that beats the public nowcast on its own target, out of sample (B-S, B1). Otherwise: inconclusive (one bounded extension) or stop for that use; if no credible source passes, B's budget is reallocated and I2 is promoted.
Months 2–3	B2 on prices, then B3 net of costs and capacity; regime and risk layers only where they change a decision; placebos	**Gate 2:** B2, paired gain over the model without the source (95% lower bound above zero) and enriched ρ above ρ^*; B3, incremental net PnL positive on both halves, capacity estimated.
Months 4–6	Isolated, logged paper run; next idea prepared; held-back window kept by Borel	**Gate 3:** paper gap at most the assumed cost plus 50%; ρ on the held-back window above ρ^*; capital only with Borel's approval.

The research investment requested now: a bounded trial of one professional Cushing dataset (a sample, then a historical trial), with delivery dates logged from the first day. After it: subscribe, change the use, or stop that source; capital only with Borel's approval. **Success at six months:** one strategy validated on held-back data with positive net PnL and an estimated capacity, or a documented stop with the gap quantified; a second idea through Gate 1; reproducible code and point-in-time data. **Stop at any time:** a source that cannot be verified point-in-time; a useful incremental gain excluded by the interval; a fusion model that does not beat the baseline; a data trial whose incremental value is below its cost; the drawdown rules.

12 Reproducibility

Python 3.11 with `requirements.txt`; `python -m pytest tests` runs 33 checks on synthetic data. Keys (ENTSO-E token, AGSI+ key) go in `.env` (template `.env.example`). HistData WTIUSD M1 zips are downloaded by the user into the folder named by `ALTDATA_HISTDATA_DIR`; `ALTDATA_PYTHON` sets the interpreter of the PowerShell driver. Test scripts check the

pre-registration hash first and some refuse to overwrite existing results: reproduce in a scratch copy and compare with `audit/compare_repro.py`.

COMMAND	OUTPUT
`python scripts/eia_wpsr_archive.py`	Release registry and first-published values
`python audit/wpsr_audits.py`	Calendar, vintage and release-time audits
`python scripts/baseline_cushing_ar2.py`	B1a baseline metrics
`python scripts/t1c_t2c_calendar.py --calendar rule`, then `--calendar archive`	Reproduction check, then T1c and T2c
`python scripts/bs0_catalog.py`; `scripts/bs_run_all.ps1`	BS0 catalogue; BS1 extraction and analysis, BS2, figures
`python scripts/b0_transmission.py --amended`; `python scripts/fig_b0.py`	B0 (amended) and its figure; `--all-years` for the post hoc sensitivity
`python scripts/a_price_proxy.py` (`--selftest`, `--dry` first)	Test AR and descriptives AH, AC (single run)
`python scripts/pnl_grid.py` (`--v1`, `--windows`)	Conditional PnL grids
`python scripts/et_reference_sensitivity.py`	Post hoc check of the Friday reference time zone (scratch recomputation of BS0 and BS2)

The full sequence, including the v1 tests T1–T5 and the satellite figures, is in `README.md`. This document describes the repository as of 26 October 2026 (revision 3); its own source, `docs/spec/spec.html`, is exported with `python docs/spec/export_spec.py`.

A References

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9. Data: U.S. Energy Information Administration; Copernicus Sentinel data 2018–2026 (ESA), modified; © OpenStreetMap contributors (ODbL); HistData.com; Open-Meteo.com (CC BY 4.0); ENTSO-E Transparency Platform; GIE AGSI+.